

QFT 2 Lecture Note

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Abstract

This is a lecture note of the QFT 2 course in EPFL

Contents

1	Lecture 1: Massless Representations of the Poincaré Group	3
1.1	Recap of UIR Construction	3
1.1.1	Helicity Basis	3
1.2	Massless Representations	3
1.2.1	Wigner Little Group	3
1.2.2	Representation of Wigner's Little Group	4
1.2.3	Interpretation of λ	5
1.2.4	Helicity Basis UIR	5
1.2.5	Values of λ	6
2	Lecture 2: Classical U(1) Gauge Theory	7
2.1	Classical Vector Field with U(1) Gauge Symmetry	7
2.1.1	Lagrangian of U(1) Gauge Theory	7
2.1.2	DoF Counting	7
2.1.3	Interaction with Other Fields	7
2.1.4	Coupling to U(1) Conserved Current	8
2.2	U(1) Gauge Field EoM (aka Maxwell's Equation)	9
3	Lecture 3: U(1) Gauge Theory Quantization	10
4	Lecture 4: Massive Vector Field	11
5	Lecture 5: Causality in QFT	12
6	Lecture 6: Scattering Theory	13
6.1	In and Out States	13
6.1.1	Free Partical States	13
6.1.2	In and Out States	13
6.2	Lippmann-Schwinger Equation	14
6.2.1	Formal Form	14
6.2.2	Asymptotic Behavior	15
6.3	S-Matrix	16
6.3.1	Definition and Properties	16
6.3.2	S Matrix Element from Lippmann-Schwinger Equation	16
6.3.3	First Born Approximation	16
7	Lecture 7: Scattering Theory II: Symmetry	17
7.1	Poincare Symmetry	17
7.1.1	Translation Invariance and Momentum Conservation	17
7.1.2	S Matrix with Full Poincare Symmetry	17
7.1.3	Generator of Symmetry	18
8	Lecture 8: Scattering Theory III: Cross Section and Decay Rate	19
8.1	Differential Cross Sections	19

8.2	DCS from S Matrix	19
8.2.1	Probability Density	19
8.2.2	Differential Event Number	19
8.2.3	Differential Cross Section	19
9	Lecture 9-11: Scattering Theory IV: Amplitudes and QED	20
10	Lecture 12: Discrete Symmetries I	20
10.1	Overview of Discrete Symmetries	20
10.1.1	Free Field Theory	20
10.1.2	Interacting Theory	20
10.2	Parity Symmetry	20
10.2.1	Definition	20
10.2.2	Faithful or Not	21
10.2.3	Scalar Field	23
	Bibliography	24

1 Lecture 1: Massless Representations of the Poincaré Group

We have classified the Unitary Irreducible Representations (UIRs) of the Poincaré group into two classes: massive representations and massless representations. In this lecture, we will focus on the massless representations.

1.1 Recap of UIR Construction

To construct a UIR of Poicare group we need to construct it for both translation and Lorentz transformation, represented as:

$$U(a) = e^{ia_\mu P^\mu}, \quad U(\Lambda) = e^{-\frac{i}{2}\omega_{\mu\nu} J^{\mu\nu}} \quad (1)$$

To do this we assume a UIR space with basis $|p, \sigma\rangle$, which diagonalizes the operator P^μ . Then we can construct the UIR of translation by:

- For Translation: $U(a) |p, \sigma\rangle = e^{ia_\mu p^\mu} |p, \sigma\rangle$.
- For Lorentz transformation we need to use **Wigner's Little Group**
 - Step 1: define a “Standard Boost” and assume it satisfy: $U(\Lambda_p) |k, \sigma\rangle = |p, \sigma\rangle$
 - Step 2: for a general lorentz transformation is given by a combination of wigner little group and the standard boost. We only need to consider a representation of the wigner little group, which is given by: $W(\Lambda, p) = \Lambda_{\Lambda p}^{-1} \Lambda \Lambda_p$.

We know that the generator of Wigner's little group is exactly given by the **Pauli-Lubanski** vector. Thus, long as we use a irreducible representation of the Wigner's little group, we can get a UIR of the Lorentz transformation.

$$\begin{aligned} U(\Lambda) |p, \sigma\rangle &= H(\Lambda_p) U(W(\Lambda, p)) |k, \sigma\rangle \\ &= \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) |\Lambda p, \sigma'\rangle \end{aligned} \quad (2)$$

1.1.1 Helicity Basis

Different Basis corresponds to different “standard boost”. For helicity basis, we choos the standard boost to be:

$$\Lambda_p^h = R(\hat{p}) e^{iK_3 \eta_p}, \quad \eta_p = \operatorname{arctanh}\left(\frac{|\vec{p}|}{p^0}\right) \quad (3)$$

which is:

- first boost to the speed of p along the z -axis, and then
- rotate to the direction of p .

Note that here K_3 is in the defining representation of Lorentz group.

1.2 Massless Representations

1.2.1 Wigner Little Group

For a massless representation, we take the standard momentum to be $k^\mu = (k, 0, 0, k)$. We can directly get the Wigner's Little group by calculating the Pauli-Lubanski vector, which is given by:

$$W^0 = -kJ^3, \quad W^1 = k(K^2 - J^i), \quad W^2 = k(-J^2 - K^1), \quad W^3 = -kJ^3 \quad (4)$$

We can see that the Wigner's Little group only have 3 generators satisfying the commutation relation of:

$$[J^3, W^1] = iW^2, \quad [J^3, W^2] = -iW^1, \quad [W^1, W^2] = 0 \quad (5)$$

This is the algebra of ISO(2) group.

1.2.2 Representation of Wigner's Little Group

We now try to find a representation for this algebra. We can recombine the generators as:

$$W^\pm = W^1 \pm iW^2 \quad (6)$$

Then the commutation relation becomes:

$$[J^3, W^+] = W^+, \quad [J^3, W^-] = -W^-, \quad [W^+, W^-] = 0 \quad (7)$$

We can see that $W^\pm$ behaves like the raising and lowering operator, but the noncommutative nature tells us that state generated by $W^\pm$ are not normalizable. We can further see this from below

To construct a representation for this algebra we can have $|k, \lambda\rangle$ as the basis of the representation space which is an eigenstate of J^3 with eigenvalue λ . Then we can see that:

$$J^3 W^+ |k, \lambda\rangle = (\lambda + 1) W^+ |k, \lambda\rangle, \quad J^3 W^- |k, \lambda\rangle = (\lambda - 1) W^- |k, \lambda\rangle \quad (8)$$

And the Casimir of the algebra is given by:

$$W^2 = -W^+ W^- \quad (9)$$

For an irreducible representation, the Casimir should be proportional to the identity operator, which means that $W^+ W^- = -c\mathbb{I}$. Moreover, we note that by definition and the restriction of representation to be unitary, we have: $W^- = (W^+)^\dagger$. Thus we can see that c should be non-negative, and we can have two cases:

- $c > 0$: then we have $(W^-)^\dagger W^- \propto 1$, which means that the lowering operator of the algebra is a unitary operator. Thus the representation space is infinite dimensional, which is not physical.
- $c = 0$: then we have $W^+ = W^- = 0$ and the representation space is one dimensional, which is physical. Thus this is the case we are interested in.

In this case, the representation of the Wigner's Little group is one dimensional and labelled by λ . The representation is given by:

$$W^\mu |k, \lambda\rangle = -\lambda k^\mu |k, \lambda\rangle \quad (10)$$

We can generalize this result to a general p by using the commutation relation of W^μ and Lorentz Generators. The results is:

Theorem 1.1

Massless Representation of Wigner's Little Group

For a massless representation of the Poincaré group, the representation of Wigner's Little group is one dimensional and labelled by λ . The representation of generators are given by:

$$W^\mu |p, \lambda\rangle = -\lambda p^\mu |p, \lambda\rangle, \quad W^\mu = -\lambda P^\mu \quad (11)$$

we can prove this by:

$$H(p) W^\mu H(p)^\dagger H(p) |k, \lambda\rangle = H(p) W^\mu |k, \lambda\rangle = -\lambda H(p) k^\mu |p, \lambda\rangle \quad (12)$$

and we have the commutation relation:

$$H(p)W^\mu H(p)^\dagger = W^\nu \Lambda_{p\nu}^\mu \quad (13)$$

1.2.3 Interpretation of λ

We can give λ a physical interpretation. We choose the standard boost as:

$$\Lambda_p^h = R(\hat{p})e^{iK_3\eta_p}, \quad \eta_p = \text{arctanh}\left(\frac{|\vec{p}|}{p^0}\right) \quad (14)$$

We know from Equation 12 that the state $|p, \lambda\rangle$ is an eigenstate of $H(p)J^3H(p)^\dagger$ eigenvector with eigenvalue λ . In the helicity basis, we plug in the form of $H(p)$ we have:

$$H(p)J^3H(p)^\dagger = R(\hat{p})e^{iK_3\eta_p}J^3e^{-iK_3\eta_p}R(\hat{p})^\dagger = R(\hat{p})J^3R(\hat{p})^\dagger = J \cdot \hat{p} \quad (15)$$

We thus see that the basis vector $|p, \lambda\rangle$ in the helicity basis is an eigenvector of the operator:

$$J \cdot \hat{p} |p, \lambda\rangle = \lambda |p, \lambda\rangle \quad (16)$$

Thus we interpret λ as the helicity of the state, which is the projection of the spin along the direction of motion.

In fact, the form of $H(p)W^\mu H(p)^\dagger$ is independent of the choice of explicit form of $H(p)$ for **massless representation**. Yet, helicity basis is good for directly showing this result.

1.2.4 Helicity Basis UIR

We explicitly write down the “standard boost” for the helicity basis:

$$\Lambda_p^h = R(\hat{p})e^{iK_3\eta_p} \quad (17)$$

where

$$e^{iK_3\eta_p} = \begin{pmatrix} \cosh(\eta_p) & 0 & 0 & \sinh(\eta_p) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh(\eta_p) & 0 & 0 & \cosh(\eta_p) \end{pmatrix}, \quad \eta_p = \text{arctanh}\left(\frac{|\vec{p}|}{p^0}\right) \quad (18)$$

and $R(\hat{p})$ is standard rotation. If we now assume that $H(p) = U(\Lambda_p^h)$ that gives us:

$$H(p) |k, \lambda\rangle = |p, \lambda\rangle \quad (19)$$

as the LT that doesn't change the λ index. Now we construct the full UIR in Helicity basis, as Equation 2, we have in Helicity basis:

$$U(\Lambda)|p, \lambda\rangle = H(\Lambda p)U(W(\Lambda, p)) |k, \lambda\rangle \quad (20)$$

and we know that $U(W(\Lambda, p))$ is generated by only one non-zero generator J^3 for the $c = 0$ case, thus, we assume the lie parameter is given by $\theta(\Lambda, p)$ which can be explicitly calculated from the definition:

$$W(\Lambda, p) = \Lambda_{\Lambda p}^{h-1} \Lambda \Lambda_p^h = e^{i\theta(\Lambda, p)J^3} \quad (21)$$

(in the defining representation of lorentz group) Then we have:

$$U(\Lambda) |p, \lambda\rangle = e^{i\theta(\Lambda, p)\lambda} |\Lambda p, \lambda\rangle \quad (22)$$

This is the representation of the Lorentz transformation for massless representation in the helicity basis.

Theorem 1.2**Massless Representation of Poincare group in Helicity Basis**

For a massless representation of the Poincaré group, the representation of the Lorentz transformation in the helicity basis is given by:

$$U(\Lambda)|p, \lambda\rangle = e^{i\theta(\Lambda, p)\lambda} |\Lambda p, \lambda\rangle \quad (23)$$

the representation of the translation is given by:

$$U(a)|p, \lambda\rangle = e^{ia_\mu p^\mu} |p, \lambda\rangle \quad (24)$$

1.2.5 Values of λ

Just as the label of massive representation the spin j can only take integer or half-integer values, the helicity λ of massless representation also have restrictions. The restriction is from the global topology of the lorentz group:

- Any 4π rotation must give back the identity element.

Thus we have:

$$e^{i4\pi\lambda} = 1, \quad \lambda \in \frac{\mathbb{Z}}{2} \quad (25)$$

Moreover, we want the state not only to be representation of **Poincare group** but also form a representation of **CPT** transformation. This shows that a single λ representation is not enough, we need to have both λ and $-\lambda$ representation to form a representation of CPT transformation.

Thus we have the final result that the helicity λ can only take integer or half-integer values, and for each λ we also need to have a $-\lambda$ representation.

For example, for photon, we have $\lambda = 1$ and $\lambda = -1$ representation.

2 Lecture 2: Classical U(1) Gauge Theory

2.1 Classical Vector Field with U(1) Gauge Symmetry

2.1.1 Lagrangian of U(1) Gauge Theory

We want to construct a theory of vector fields, but more than that we require the theory to have a U(1) gauge symmetry, which means that:

$$A_{\mu(x)} \rightarrow A_{\mu(x)} - \partial_{\mu}\alpha(x) \quad (26)$$

is a local symmetry of the theory for any function $\alpha(x)$. We can show that the only Lagrangian that is invariant under this gauge symmetry is:

Theorem 2.1

Lagrangian of U(1) Gauge Theory

The Lagrangian of a U(1) gauge theory is given by:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad (27)$$

where $F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$ is the field strength tensor.

2.1.2 DoF Counting

We now count the degree of freedom of the theory. Naively, we can see A_{μ} is a real 4-vector, which has 4 DoF. However, we have a gauge symmetry, which means that we can use the gauge symmetry to fix one of the DoF. Thus, we have 3 DoF left.

This can also be seen if we view the field strength tensor $F_{\mu\nu}$ as the fundamental DoF of the theory. We can see that $F_{\mu\nu}$ is an antisymmetric 2-tensor, which has 6 independent components. However, we have the Bianchi identity:

$$\varepsilon^{\mu\nu\rho\sigma}\partial_{\nu}F_{\rho\sigma} = 0 \quad (28)$$

which gives finally 3 independent DoF for the field strength tensor, which is consistent with the DoF counting from the gauge field A_{μ} .

2.1.3 Interaction with Other Fields

We now consider all possible interactions between the gauge field A_{μ} and other fields.

- Choice 1: $F_{\mu\nu}$ coupling, first we can consider following interaction:

$$F_{\mu\nu}F^{\mu\nu}\varphi, \quad F_{\mu\nu}\bar{\psi}\sigma^{\mu\nu}\psi \quad (29)$$

However, these interaction have an explicit $F_{\mu\nu}$, one can prove that these are non-renormalizable.

- Choice 2: A_{μ} coupling. Yet we may face the problem that A_{μ} is not gauge invariant, thus we need to define a gauge transformation for the other fields to make the interaction gauge invariant. There are many ways to do this:
 - Dirac Field: we can define the gauge transformation for a Dirac field ψ as:

$$\psi(x) \rightarrow e^{iq\alpha(x)}\psi(x) \quad (30)$$

- Complex Scalar Field: we can define the gauge transformation for a complex scalar field φ as:

$$\varphi(x) \rightarrow e^{iq\alpha(x)}\varphi(x) \quad (31)$$

- Real Scalar Field: we can define the gauge transformation for a real scalar field φ as:

$$\varphi(x) \rightarrow \varphi(x) + M\alpha(x) \quad (32)$$

With these gauge transformation, we can construct the following gauge invariant interaction with a tool called covariant derivative:

Definition 2.2

Covariant Derivative

The covariant derivative is defined as a derivative transformed under the gauge transformation as:

$$D_\mu \rightarrow U(x)D_\mu U(x)^{-1} \quad (33)$$

for U(1) gauge theor, we have $U(x) = e^{iq\alpha(x)}$, thus we have:

$$D_\mu = \partial_\mu + iqA_\mu \quad (34)$$

Thus we can couple by using the covariant derivative, for example:

$$\bar{\psi}\gamma^\mu D_\mu \psi, \quad (D_\mu \varphi)^* D^\mu \varphi \quad (35)$$

2.1.4 Coupling to U(1) Conserved Current

We then can see an interesting fact if we single out the interaction term, we can see that for the Dirac field, we have:

$$\mathcal{L}_{\text{int}} = -q\bar{\psi}\gamma^\mu A_\mu \psi \quad (36)$$

and for the complex scalar field, we have:

$$\mathcal{L}_{\text{int}} = -iq(\varphi^* \partial^\mu \varphi - \varphi \partial^\mu \varphi^*) A_\mu + q^2 A_\mu A^\mu \varphi^* \varphi \quad (37)$$

We may see a fact that **the U(1) gauge field couple to the global U(1) current**. This is not a coincidence. In fact, in we want to lift a global U(1) symmetry to a local U(1) symmetry, we know that the variation of the matter field Lagrangian under the global U(1) transformation is given by:

$$\delta \mathcal{L}_m = J_N^\mu \partial_\mu \alpha(x) \quad (38)$$

Then if we want the full Lagrangian with the gauge field to be invariant under the local U(1) transformation, we need to add a term to cancel the variation. We assume that the full Lagrangian is given by:

$$\mathcal{L} = \mathcal{L}_m + \mathcal{L}_{\text{gauge}} + A^\mu J_\mu \quad (39)$$

Then:

$$\delta \mathcal{L} = J_N^\mu \partial_\mu \alpha(x) + A^\mu \delta J_\mu + \delta A^\mu J_\mu = 0 \quad (40)$$

we know that $\delta A^\mu = -\partial^\mu \alpha(x)$, thus if $\delta J_\mu = 0$ under the gauge transformation (Dirac Field Case), we can see that the symmetry condition gives us:

$$J_N^\mu = J^\mu \quad (41)$$

Thus we can interpret the global U(1) symmetry conserved current as **electric current**.

Remark

Note that $\delta I_\mu = 0$ is only true for dirac field. For scalar field, the current is not invariant under the gauge transformation, thus we have $\delta J_\mu \neq 0$, which cause to the extra term.

2.2 U(1) Gauge Field EoM (aka Maxwell's Equation)

3 Lecture 3: $U(1)$ Gauge Theory Quantization

4 Lecture 4: Massive Vector Field

5 Lecture 5: Causality in QFT

6 Lecture 6: Scattering Theory

Here we basically follow Weinberg's approach to scattering theory. In fact the correct understanding of this should be based on the LSZ reduction formula, but we will not go into that in this lecture.

6.1 In and Out States

To set up the scattering theory, we need to define the in and out states.

6.1.1 Free Partical States

In free QFT, we can define the one-particle or multi-particle states as the eigen state of P^μ with a series of labels $\{\alpha\}$, these states are:

$$P^\mu |\phi_\alpha\rangle = p_\alpha^\mu |\phi_\alpha\rangle \quad (42)$$

A wave packet state can be defined as a superposition of these states:

$$|\phi\rangle = \int d^3\alpha g(\alpha) |\phi_\alpha\rangle \quad (43)$$

6.1.2 In and Out States

For an interacting QFT, we can define the in and out states as being asymptotically free, which means that they can be approximated by free particle states at $t \rightarrow \pm\infty$. Thus we have:

Definition 6.1

In and Out States

The in and out states are defined as being asymptotically free

- **In states:** $|\psi_\alpha^+\rangle$ behaves as $|\phi_\alpha\rangle$ when $t \rightarrow -\infty$.
- **Out states:** $|\psi_\alpha^-\rangle$ behaves as $|\phi_\alpha\rangle$ when $t \rightarrow +\infty$.

Here are some general properties of the in and out states:

1. **Moller Operators:** We can define in and out state quantitatively by the Moller operators:

$$\Omega_\pm = \lim_{t \rightarrow \mp\infty} \exp(iH(t-t_0)) \exp(-iH_0(t-t_0)) \quad (44)$$

where H is the Hamiltonian of the interacting theory and H_0 is the Hamiltonian of the free theory. Then we have:

$$|\psi_\alpha^\pm\rangle = \Omega_\pm |\phi_\alpha\rangle \quad (45)$$

2. **Moller Operator Commutation** The Moller operators commute with the full Hamiltonian H :

$$H\Omega_\pm = \Omega_\pm H_0 \quad (46)$$

3. **Energy Eigen States** The in and out states are eigen states of the full Hamiltonian H with eigenvalue E_α :

$$H |\psi_\alpha^\pm\rangle = E_\alpha |\psi_\alpha^\pm\rangle \quad (47)$$

where E_α is the energy of the state $|\phi_\alpha\rangle$.

4. **Normalization** we assume Moller operators are unitary, which means that the in and out states are normalized as the free particle states:

$$\langle\psi_\alpha^\pm|\psi_\beta^\pm\rangle = \langle\phi_\alpha|\phi_\beta\rangle = \delta(\alpha - \beta) \quad (48)$$

Remark

We use the notation $\delta(\beta - \alpha)$ for the normalization of the states, which should be understood as the combination of the relativistic delta function and the Kronecker delta function.

6.2 Lippmann-Schwinger Equation

6.2.1 Formal Form

We want to derive a formal solution for the in and out states. The result is the Lippmann-Schwinger equation:

Theorem 6.2

Lippmann-Schwinger Equation

The in and out states can be formally expressed as:

$$|\psi_\alpha^\pm\rangle = |\phi_\alpha\rangle + \frac{H_I}{E_\alpha - H_0 \pm i\varepsilon} |\psi_\alpha^\pm\rangle \quad (49)$$

- Derivation 1: Matrix Inversion

This can be proven by the fact that the in and out states are eigen states of the full Hamiltonian H with eigenvalue E_α :

$$H |\psi_\alpha^\pm\rangle = E_\alpha |\psi_\alpha^\pm\rangle \Rightarrow (H_0 - E_\alpha) |\psi_\alpha^\pm\rangle = -H_I |\psi_\alpha^\pm\rangle \quad (50)$$

Plugging in the ansatz of $|\psi_\alpha^\pm\rangle = a|\phi_\alpha\rangle + \Delta|\psi_\alpha^\pm\rangle$ and rearranging the terms, we can get the Lippmann-Schwinger equation.

- Derivation 2: Moller Operators

Another QM style derivation is to use the Moller operators. We can use the derivative and integral trick to prove that:

$$(\Omega_\pm - 1) |\phi_\alpha\rangle = \frac{H_I}{E_\alpha - H_0 \pm i\varepsilon} \Omega_\pm |\phi_\alpha\rangle \quad (51)$$

Thus gives us the Lippmann-Schwinger equation. In this derivation we need to explicitly add the $\pm i\varepsilon$ term for in and out states to make the integral of time converge. There is a sign difference because the in and out states are defined at $t \rightarrow \mp\infty$.

Apart from the formal form we can plug in a complete bases of free particle states to get a more explicit form of the Lippmann-Schwinger equation:

$$I = \int d\alpha |\phi_\alpha\rangle \langle \phi_\alpha| \quad (52)$$

Remark

This complete condition is a good assumption for any lorentz symmety theoretical Hilbert Space.

Thus we have:

Theorem 6.3**Lippmann-Schwinger Equation with Free Particle Bases**

The in and out states can be formally expressed as:

$$|\psi_\alpha^\pm\rangle = |\phi_\alpha\rangle + \int d\beta \frac{1}{E_\alpha - E_\beta \pm i\varepsilon} \langle\phi_\beta|H_I|\psi_\alpha^\pm\rangle |\phi_\beta\rangle \quad (53)$$

we often call the term $T_{\beta\alpha}^\pm = \langle\phi_\beta|H_I|\psi_\alpha^\pm\rangle$ the T-matrix element, which is the transition amplitude from the free particle state $|\phi_\beta\rangle$ to the in or out state $|\psi_\alpha^\pm\rangle$.

6.2.2 Asymptotic Behavior

We mainly care about the asymptotic behavior of the in and out states. With the Lippmann-Schwinger equation, we can explicitly see that:

Remark**Heisenberg Picture**

All calculation in the scattering theory is based on Heisenberg picture. We don't say we evolve a state in time, but we say the asymptotic behavior of the state is like this or that, which is the behavior of the state:

$$e^{-iHt} |\psi\rangle \quad (54)$$

Please understand this as acting an operator on the time independent state, not as evolving the state in time.

In scattering theory, we prefer to use a wave packet or we can't physically "localize" a state as a particle. Thus we have:

$$\begin{aligned} |\psi^\pm\rangle &= \int d\alpha g(\alpha) |\psi_\alpha^\pm\rangle \\ &= \int d\alpha g(\alpha) \left(|\phi_\alpha\rangle + \int d\beta \frac{1}{E_\alpha - E_\beta \pm i\varepsilon} T_{\beta\alpha}^\pm |\phi_\beta\rangle \right) \end{aligned} \quad (55)$$

Here we focus on the in state's asymptotic behavior, the out state is quite the same. Consider the asymptotic state of $|\psi^+\rangle$:

$$\begin{aligned} e^{-iHt} |\psi^+\rangle &= \int d\alpha g(\alpha) e^{-iE_\alpha t} |\psi_\alpha^+\rangle \\ &= \int d\alpha g(\alpha) e^{-iE_\alpha t} |\phi_\alpha\rangle + \int d\alpha g(\alpha) e^{-iE_\alpha t} \int d\beta \frac{1}{E_\alpha - E_\beta + i\varepsilon} T_{\beta\alpha}^+ |\phi_\beta\rangle \end{aligned} \quad (56)$$

- $t \rightarrow -\infty$ Behavior

When $t \rightarrow -\infty$, we want to integrate out the E_α variable, to do this we have to use the complex integral of **Upper Half Plane** to make $e^{-iE_\alpha t}$ converge. Thus the second term vanishes for there are no poles in UHP and we have:

$$e^{-iHt} |\psi^+\rangle = \int d\alpha g(\alpha) e^{-iE_\alpha t} |\phi_\alpha\rangle \quad (57)$$

which is exactly the definition of the in state.

- $t \rightarrow +\infty$ Behavior

When $t \rightarrow +\infty$, we want to integrate out the E_α variable, to do this we have to use the complex integral of **Lower Half Plane** to make $e^{-iE_\alpha t}$ converge. Thus we have:

$$e^{-iHt} |\psi^+\rangle = \int d\alpha g(\alpha) e^{-iE_\alpha t} \left(|\phi_\alpha\rangle - 2\pi i \int d\beta \delta(E_\alpha - E_\beta) T_{\beta\alpha}^+ |\phi_\beta\rangle \right) \quad (58)$$

6.3 S-Matrix

Finally we can define the S-matrix as the observable of the scattering process.

6.3.1 Definition and Properties

The S-matrix element is defined as;

Definition 6.4

S-Matrix Element

The S-matrix element is defined as the transition amplitude from an in state to an out state:

$$S_{\beta\alpha} = \langle \psi_\beta^- | \psi_\alpha^+ \rangle \quad (59)$$

Here are some basic properties of the S-matrix:

- **Non interacting limit** In the non-interacting limit, the in and out states are the same, thus we have:

$$S_{\beta\alpha} = \delta(\beta - \alpha) \quad (60)$$

- **Unitarity** The S-matrix is unitary, which means that:

$$\int d\beta S_{\beta\alpha}^* S_{\beta\gamma} = \delta(\alpha - \gamma) \quad (61)$$

- **Moller Operators** The S-matrix can be expressed in terms of the Moller operators as:

$$S = \Omega_-^\dagger \Omega_+ \quad S_{\beta\alpha} = \langle \phi_\beta | \Omega_-^\dagger \Omega_+ | \phi_\alpha \rangle \quad (62)$$

6.3.2 S Matrix Element from Lippmann-Schwinger Equation

With the Lippmann-Schwinger equation, we can express the S-matrix element in terms of the T-matrix element, we play a trick of inserting an identity operator in the middle:

$$S_{\beta\alpha} = \langle \psi_\beta^- | \psi_\alpha^+ \rangle = \langle \psi_\beta^- | e^{iHt} e^{-iHt} | \psi_\alpha^+ \rangle \quad (63)$$

If we take the limit of $t \rightarrow +\infty$ and use the asymptotic behavior of the in and out states, we can get:

$$S_{\beta\alpha} = \delta(\alpha - \beta) - 2\pi i \delta(E_\alpha - E_\beta) T_{\beta\alpha}^+ \quad (64)$$

One then also can see that the asymptotic behavior of the in and out states is related to the S-matrix element:

$$|\psi_\alpha^+\rangle \sim \int d\beta S_{\beta\alpha} |\phi_\beta\rangle \quad (65)$$

6.3.3 First Born Approximation

The first Born approximation is to approximate the T-matrix element by:

$$T_{\beta\alpha}^+ \sim \langle \phi_\beta | H_I | \phi_\alpha \rangle = V_{\beta\alpha} \quad (66)$$

This gives us a simple approximation of the S-matrix element:

$$S_{\beta\alpha} \sim \delta(\beta - \alpha) - 2\pi i \delta(E_\alpha - E_\beta) V_{\beta\alpha} \quad (67)$$

7 Lecture 7: Scattering Theory II: Symmetry

This section we want to talk about the symmetry of the scattering theory, which is a very important topic. For a symmetry covariant quantum theory, every symmetry should be related to a Unitary operator:

$$U \tag{68}$$

For a scattering theory, when we say it is some symmetry invariant, we mean that:

Definition 7.1

Symmetry of Scattering Theory

The symmetry operator U act on the asymptotic states as the corresponding symmetry operator of the free theory act on free particle states, which means that:

$$U|\psi_\alpha^\pm\rangle \sim U_0|\phi_\alpha\rangle \tag{69}$$

Notice that this is a stronger condition that requiring the theory to have some symmetry.

This section we will focus on, if we assume that the scattering theory has some symmetry (which should be the case for a theory we want). What are the consequences of this symmetry?

7.1 Poincare Symmetry

We focus on scalar field theory for simplicity, thus the free theory states are labeled by momentums $|\{p_i\}\rangle$, and we use the notation $|\{k_i\}^\pm\rangle$ for asymptotic states. The poincare symmetry is generated by $U(\Lambda, a)$. With a theory of this symmetry, we have:

$$\begin{aligned} U_0(\Lambda, a)|\{p_i\}\rangle &= e^{i\sum_i p_i^\mu a_\nu \Lambda_\mu^\nu} |\{\Lambda p_i\}\rangle \\ U(\Lambda, a)|\{k_i\}^\pm\rangle &= e^{i\sum_i k_i^\mu a_\nu \Lambda_\mu^\nu} |\{\Lambda k_i\}^\pm\rangle \end{aligned} \tag{70}$$

Here let's focus on its consequences.

7.1.1 Translation Invariance and Momentum Conservation

We plug in a $U(\Lambda, a)$ in the definition of the S matrix element:

$$S(\{p_i\}, \{k_i\}) = e^{i\sum_i p_i^\mu a_\nu \Lambda_\mu^\nu} e^{-i\sum_i k_i^\mu a_\nu \Lambda_\mu^\nu} S(\{\Lambda p_i\}, \{\Lambda k_i\}) \tag{71}$$

The LHS is independent of a , if we take $\Lambda = I$ thus we must have:

$$\sum_i p_i^\mu = \sum_i k_i^\mu \tag{72}$$

This shows that:

- The Translation Invariance of the scattering theory implies the conservation of the total momentum.

Thus the S matrix element can be written as a stronger form in a translational invariant theory:

$$S_{\beta\alpha} = \delta_{\alpha\beta} - (2\pi)^4 \delta^{(4)}(P_f - P_i) i\mathcal{M}_\beta \tag{73}$$

where P_f and P_i are the total momentum of the final and initial states, and $\mathcal{M}_\beta$ is called the **scattering amplitude**.

7.1.2 S Matrix with Full Poincare Symmetry

Then we can further plug in full poincare transformation, we have:

$$[S, U_0(\Lambda, a)] = 0 \tag{74}$$

The deduction is straightforward from definition. We can define the generators of the Poincare group in the free and full interacting theory as:

$$\begin{aligned} G_0 &= (H_0, P_0, J_0, K_0) \\ G &= (H, P, J, K) \end{aligned} \tag{75}$$

then we simply have:

$$[S, G_0] = 0 \tag{76}$$

In fact, we can further deduce a relation between the generators and Moller operators:

$$G\Omega_{\pm} = \Omega_{\pm}G_0 \tag{77}$$

7.1.3 Generator of Symmetry

In fact in a **Typical Interacting Theory**, the generators of the symmetry in the full interacting theory are in the following relation with the generators in the free theory:

$$H = H_0 + H_I \quad P = P_0 \quad J = J_0 \quad K = K_0 + \Delta K \tag{78}$$

Now if we assume this holds, we can further deduce some results:

$$[J_0, H_I] = [P_0, H_I] = 0 \tag{79}$$

and for boost generator, we have:

$$[K, H_I] + [\Delta K, H_0] = 0 \tag{80}$$

8 Lecture 8: Scattering Theory III: Cross Section and Decay Rate

Now we focus on how to compute some physical quantities from the S matrix of a Scattering Theory. This discussion mainly follows the approach in Peskin and Schroeder's book [1].

8.1 Differential Cross Sections

8.2 DCS from S Matrix

Now we discuss how to get

8.2.1 Probability Density

An naive observable is the probability of event happening. To describe probability in QM we use a **projection measuring**:

$$P(\psi \rightarrow \Omega) = \langle \psi | \mathbb{P}_\Omega | \psi \rangle \quad (81)$$

In QFT we mainly care about the probability of a final state being generated under given initial condition of the scattering. A final state (for example for spinless particles) can be described as a free many particle state with certain momenta and spins:

$$|p_1, p_2, \dots, p_n\rangle \quad (82)$$

Then for exampl

In reality, our scattering set up may not be able to fix

8.2.2 Differential Event Number

8.2.3 Differential Cross Section

9 Lecture 9-11: Scattering Theory IV: Amplitudes and QED

YL: [These sections discussed about the scattering amplitudes from Feymann Diagrams and QED, Form my perspective, it is better to understand them from LSZ reduction formula, and it is incomplete to study them without mentioning renormalization. Thus, I'll not wite a note on these sections.]

10 Lecture 12: Discrete Symmetries I

10.1 Overview of Discrete Symmetries

There are 3 discrete symmetries important in QFT: C, P, T .

10.1.1 Free Field Theory

In free field theory, most theory preserve all C, P, T symmetry. However, not Weyl Fermion. First, in Weyl fermion C and P , must go together, remember in analysis of adjoint representation, we have seen that:

$$P : \left(\frac{1}{2}, 0\right) \rightarrow \left(0, \frac{1}{2}\right) \quad (83)$$

For a Weyl fermion, to have it represent a P symmetry, we need to define the symmetry transformation as:

$$P : \psi_L(x) \rightarrow \varepsilon \psi_L^*(Px) \quad (84)$$

This automatically does a C transformation at the same time, due to the fact that the charge change from 1 to -1 when we take the complex conjugate. So we have:

$$\psi_L \rightarrow e^{iq} \psi_L \quad \varepsilon \psi_L^* \rightarrow e^{-iq} \varepsilon \psi_L^* \quad (85)$$

10.1.2 Interacting Theory

Generally, interactions break some of the discrete symmetries. However, they never break CPT symmetry, which is a fundamental theorem in QFT.

Theorem 10.1

CPT Theorem

In any local, (P,O)Lorentz invariant quantum field theory, will always have an exact CPT symmetry:

$$\Theta = CPT \quad (86)$$

Experimentally, this is a fact:

- In standard model, both P and T are broken.
- In QED C and P exist.
- In QCD and standard model strong interaction, preserves P and T . This is a puzzle called the strong CP problem, why nature chooses to preserve this symmetry.

10.2 Parity Symmetry

10.2.1 Definition

- Non-relativistic Classical and Quantum Mechanics

Classically, we can define parity as a transformation on the canonical variables, it is a symmetry when preserving the dynamics of the system.

$$P : x^i \rightarrow -x^i, p^i \rightarrow -p^i \quad (87)$$

we call object transform like this a vector and object transform like:

$$P : L^i \rightarrow L^i, S^i \rightarrow S^i \quad (88)$$

as a pseudovector. Now we perform a canonical quantization. As Wigner theorem tells us, the symmetry transformation are realized as unitary or anti-unitary operators on the Hilbert space. For parity operator, we can define a unitary operator U_P such that:

$$U_P^\dagger x^i U_P = -x^i, \quad U_P^\dagger p^i U_P = -p^i \quad (89)$$

• Relativistic Quantum Field Theory

Now we generalize this discussion to relativistic quantum field theory. We define the parity transformation as a transformation on the spacetime coordinates:

$$P : x^\mu \rightarrow P^\mu{}_\nu x^\nu, \quad P^\mu{}_\nu = \text{diag}(1, -1, -1, -1) \quad (90)$$

Notice that the matrix satisfies:

$$P^\mu{}_\rho P^\rho{}_\sigma = \delta^\mu_\sigma \quad (91)$$

As a analogue of the continuous symmetry we discussed before.

- In QFT, the symmetry is realized by a unitary operator U_P on the Hilbert space. And forms a Unitary representation of the parity transformation, with suitable commutation relations with other symmetry operators.

$$U_P^\dagger P^\mu U_P = P^\mu{}_\nu P^\nu, \quad U_P^\dagger J^{\mu\nu} U_P = P^\mu{}_\rho P^\nu{}_\sigma J^{\rho\sigma} \quad (92)$$

- Usually the Hilbert space can be labeled by momentum and spin and other quantum numbers. Due to the commutation relation of the momentum operator, angular momentum operator and the parity operator, we have:

$$U_P |\{p_i\}\{\sigma_i\}\rangle = \prod_i \eta_i |\{P^\mu{}_\nu p_i^\nu\}\{\sigma_i\}\rangle \quad (93)$$

Note here we don't consider other quantum number that may transform under parity transformation.

- Dynamical fields transform under the parity transformation as:

$$U_P^\dagger \varphi_a(x) U_P = \hat{P}_{ab} \varphi_b(Px) \quad (94)$$

where $\hat{P}$ is a representation of the parity transformation on the field space.

10.2.2 Faithful or Not

In fact we have the degree of freedom to choose how field transform under the parity transformation, thus $\hat{P}_{ab}$ can be chosen. Of course, there are some standards, we always would like the parity to be a symmetry for the theory (of course, may fail!) and the presentation to the faithful (of course may fail too).

Now let's consider, when will we fail to have a faithful representation of the parity transformation on a QFT. This is the case of **Majorana fermion**.

Generally, the parity transformation operator U_P is a **faithful representation** if $U_P^2 = 1$. In term of the acting on the fields, we have:

$$(U_P^2)^\dagger \varphi_a(x) (U_P^2) = \hat{P}_{ab}^2 \varphi_b(x) = \varphi_a(x)$$

$$U_P^2 |\{p_i\}\{\sigma_i\}\rangle = \prod_i \eta_i^2 |\{p_i\}\{\sigma_i\}\rangle = |\{p_i\}\{\sigma_i\}\rangle \quad (95)$$

Thus we have $\hat{P}^2 = 1$ and $\eta_i^2 = 1$.

We observe that U_P^2 is a symmetry operator of a internal symmetry. As we have discussed, this internal symmetry can be chosen quite arbitrarily for certain field.

- **If it is a Lie Group Symmetry**

Then we have:

$$U_P^2 = e^{i\alpha_a Q_a} \quad (96)$$

Where Q_a is the generator of the internal symmetry. Then we can redefine the parity transformation as:

$$U_{P'} = U_P e^{-i\alpha_a \frac{Q_a}{2}} \quad (97)$$

Then it gives us a faithful representation of parity transformation, if we redefine the parity transformation from U_P to $U_{P'}$:

$$U_{P'}^2 = U_P e^{-i\alpha_a \frac{Q_a}{2}} U_P e^{-i\alpha_a \frac{Q_a}{2}} = U_P^2 = 1 \quad (98)$$

Note

Note that the internal symmetry generator should commute with a spactime symmetry, for they are naturally different.

- **If it is a Discrete Symmetry** (Majorana Fermion)

However, sometimes parity transformation for some field may limit U_P^2 to be a discrete symmetry and representation will never be faithful. This is the case of Majorana fermion, where we have:

$$U_P^2 = (-1)^F \quad (99)$$

In this case, we can never redefine. Of course, one may think that $(-1)^F = e^{i2\pi J^3}$, yet J^3 is a spactime symmetry, the commutation may not be guaranteed.

In Majorana Fermions, the mass term gives us:

$$m\psi_L^T \varepsilon \psi_L + h.c. \quad (100)$$

If we want the mass term to preserve the parity symmetry, we need to have:

$$U_P : \psi_L(x) \rightarrow \eta \varepsilon \psi_L^*(Px) \quad \eta^2 = -1 \quad (101)$$

This lead to a fact that:

$$U_P^2 : \psi_L(x) \rightarrow -\psi_L(x) \quad (102)$$

There is a statement that:

Theorem 10.2

Besides Majorana Fermion, all other fields can be given a good faithful representation of which the parity transformation is.

Thus now we focus on the faithful representation case. In this case, one can prove that the field will always transform as:

$$U_P^\dagger \varphi_a(x) U_P = \eta_a \varphi_a(Px) \quad \eta_a^2 = 1 \quad (103)$$

where η_a is called the intrinsic parity of the field.

10.2.3 Scalar Field

Consider a faithful representation of parity transformation on a scalar field. We have two cases:

$$U_P^\dagger \varphi(x) U_P = \varphi(Px) \quad U_P^\dagger \varphi(x) U_P = -\varphi(Px) \quad (104)$$

The first case is called scalar field, the second case is called pseudoscalar field.

- **Clein Gordon Field**

Focus on CG field, we can explicitly calculate how parity act on the creation and annihilation operator and so on all operators. By assuming:

$$U_P^\dagger \varphi(x) U_P = \eta \varphi(Px) \quad \eta^2 = 1 \quad (105)$$

we can have:

$$U_P^\dagger a_k U_P = \eta_P a_{k_P} \quad U_P^\dagger a_k^\dagger U_P = \eta_P a_{k_P}^\dagger \quad (106)$$

This tells us the vacuum is invariant under parity transformation, and we can also derive the transformation of conserved currents, which exactly match the adjoint representation of parity on poincare algebra.

- **Complex Scalar Field**

For a complex scalar field, we ordinarily define the parity transformation as:

$$U_P^\dagger \phi(x) U_P = \eta_P \phi(x_P) \quad U_P^\dagger \phi^*(x) U_P = \eta_P \phi^*(x_P) \quad (107)$$

Bibliography

- [1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*. Reading, Mass: Addison-Wesley Pub. Co, 1995.